Michael Yang

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Education

Carleton University Ottawa, ON
Bachelor of Computer Science, Honours in AI and ML, Co-op Option *Sept. 2023 – Apr. 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Discrete Structures II, Systems Programming, Software Engineering

Technical Skills

Languages: Java, Python, C, C++, TypeScript, JavaScript, HTML/CSS, Lua
Frameworks/Libraries: React, Next.js, Node.js, Express.js, Tailwind CSS, pytest, DeepEval, LangChain, Agno, Convex
Developer Tools: Git, GitHub, GitHub Actions, Docker, Kubernetes, k6, MongoDB, LanceDB, Pinecone, Prometheus, Grafana

Experience

Software Developer Co-op May 2026 – Aug 2026
JSI Telecom *Ottawa, ON*

- Implemented k6 load tests against the LLM inference service, simulating **50+** concurrent virtual users and **80,000+** requests to measure model latency and throughput.
- Integrated the benchmark suite into GitHub Actions and visualized latency and throughput on Grafana to compare serving configurations across CI runs.
- Diagnosed bottlenecks erroring out **74%** of requests to the LLM inference microservice under **8-million-entry** k6 load from in-cluster Kubernetes pods by correlating Grafana metrics with pod logs; scaled replicas via kubectl to cut errors to **<1%**.

Software Developer Co-op Sept. 2025 – Dec. 2025
JSI Telecom *Ottawa, ON*

- Developed task-specific LLM agents in Python using **LanceDB** and Agno, including an FAQ agent and a prompt-alignment verifier.
- Maintained **95%+** output consistency with DeepEval, wrote pytest unit and integration tests reaching **90%+** code coverage, and added MkDocs documentation for the evaluation harness.
- Implemented a custom HTTP and gRPC client wrapper to instrument **500+** daily outbound requests with Prometheus and Grafana, reducing mean time to resolution (MTTR) by **60%**.

Incubation Developer Co-op June 2025 – Aug. 2025
Nokia *Ottawa, ON*

- Developed a Python simulation tool to model and validate a 5G traffic-management proof of concept, simulating **7000+** concurrent packet distributions to identify bottleneck flows.
- Implemented parameterized pytest suites for packet-distribution scenarios to provide repeatable validation of the incubation proof of concept.

Leadership

Founder & Lead Developer, Independent Game Studio | Luau, Roblox Studio July 2026 – Present

- Founded an independent Roblox studio and led a team of 3D modelers, builders, QA testers, and a co-developer to ship a multiplayer survival-horror game that reached **10M+** visits and **14,000+** peak concurrent users (CCU) within one month.
- Designed core gameplay systems and NPC AI in **Luau** using finite state machines and pathfinding, and continued leading the development of those systems after launch.

Projects

CanadaChain | **3rd Place, Ottawa Red Team Hackathon** | TypeScript, Docker, Next.js, Tailwind CSS, Convex, React

- Developed a full-stack cryptographic supply-chain provenance platform in TypeScript, Next.js, Convex, and React, including a public verification interface and a backend that computes Canadian content and Made in Canada / Product of Canada designations.
- Implemented tamper-evident attestations with **SHA-256** content hashes and **Ed25519** supplier signatures, enabling independent end-to-end verification of provenance records.

AIsk | JavaScript, React, LangChain, Tailwind CSS, Pinecone, Express.js, MongoDB, Tavily, Node.js

- Developed a MERN retrieval-augmented generation chatbot that queries Healthline through **Tavily**, embeds document chunks in **Pinecone** (top-5 matches, similarity ≥ 0.8), and generates responses with LangChain and Ollama.
- Stored user accounts and conversation history in MongoDB and scoped Tavily retrieval to Healthline as the document source.